

```

1  LIBRARY ieee;
2  USE ieee.std_logic_1164.ALL;
3  USE ieee.numeric_std.ALL;
4
5  ENTITY s22_3a IS
6      PORT (
7          clk : IN STD_LOGIC;
8          y : OUT STD_LOGIC_VECTOR(3 DOWNTO 0)
9      );
10 END ENTITY;
11
12 ARCHITECTURE behavioral OF s22_3a IS
13     SIGNAL q : STD_LOGIC_VECTOR(3 DOWNTO 0) := x"6";
14     SIGNAL a, b, c, d, e : STD_LOGIC;
15 BEGIN
16
17     --flipflops
18     ff : PROCESS (clk)
19     BEGIN
20         IF rising_edge(clk) THEN
21             q(0) <= a;
22             q(1) <= b;
23             q(2) <= c;
24             q(3) <= d;
25         END IF;
26     END PROCESS ff;
27
28     --combinational process
29     a <= (NOT q(3)) AND e;
30     b <= q(0) AND e;
31     c <= q(1) AND e;
32     d <= q(2) AND e;
33     e <= NOT (q(3) AND q(3)) AND q(2) AND q(1) AND (NOT q(0));
34
35     --the output
36     y <= q;
37
38 END ARCHITECTURE behavioral;

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